GUNJAN RAJGURE

AI/ML ENGINEER | LLM SYSTEMS | BIOMEDICAL AI | DATA SCIENCE

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PROFESSIONAL SUMMARY

AI/ML engineer and biomedical researcher with **three years** of enterprise systems experience and advanced work spanning **multimodal representation learning**, **biomedical NLP**, **agentic RAG**, **distributed LLM serving**, **statistical modeling**, APIs, data engineering, and production-oriented evaluation. Builds **reproducible systems** that connect model behavior to measurable performance, grounded evidence, and usable applications.

PROFESSIONAL EXPERIENCE

Research Assistant | University of Delaware | Newark, DE May 2025 - Present

- Architected **PhenoZebra**, an end-to-end multimodal representation-learning pipeline indexing **69,003 clinical cases** across **1,727 rare diseases** and **41,954 image-linked cases** from **37 medical departments**.
- Built a **GPU-accelerated** feature stack using **BioClinical-ModernBERT**'s **8,192-token context** and **MedGemma/SigLIP** vision representations, generating **69,003 text embeddings** and **1,464 multimodal disease fingerprints**.
- Optimized **NVIDIA A100 80GB** inference through **bfloat16** precision, batched generation, reusable embedding caches, and checkpoint recovery, mapping **175,659 symptoms** to **HPO** at **60.55 terms per second** in approximately **49 minutes**.
- Developed a vectorized similarity engine across **1,464 diseases** using **cosine similarity** and ontology-based **Jaccard constraints**, surfacing **711,264** low-overlap candidates including **614,133 cross-department relationships**.
- Designed explainability components with **UMAP**, **NetworkX**, **evidence-linked metadata**, and ranked outputs, producing reproducible **PyTorch**, **JSON**, **CSV**, and visual artifacts connected to clinical narratives and source literature.
- Operationalized **eMIND biomedical NLP** through dependency parsing, **named-entity recognition**, normalization, masked-argument **BERT relation extraction**, rule inference, negation detection, and ontology mapping at approximately **0.94 precision** and **0.84 recall**.
- Developed **eMIND Visuals**, an **interactive AI application** supporting search across genes, variants, **UniProt** accessions, and **PubMed** IDs with structured relationships and **evidence highlighted** within source abstracts.
- Engineered scalable **Python** pipelines for **iTextMine/PMC** using single-pass **JSONL** processing, **PMCID normalization**, **deduplication**, multi-destination routing, malformed-record handling, **throughput monitoring**, and reproducible corpus generation.

Systems Engineer | Tata Consultancy Services | Pune, India July 2021 - July 2024

- Led an **eight-member team** delivering an **NLP classification** system across **100,000+** text records, achieving **95% accuracy** through **validation** and iterative error analysis.
- Developed **machine-learning** and **statistical workflows** for text classification, **feature engineering**, **model validation**, and data-driven decision support using **Python** and **scikit-learn**.
- Designed and maintained enterprise full-stack applications and **50+ REST APIs** with **Keycloak authentication and authorization** for secure, maintainable client solutions.
- Automated **model evaluation**, scoring, analysis, and reporting with **Python**, reducing manual effort by **60%** and improving **repeatability** across engineering workflows.
- Translated ambiguous **client requirements** into **technical designs**, **documented workflows**, and maintainable releases while collaborating across **Agile** engineering and stakeholder teams.
- Led internal technology initiatives, coordinated an **eight-member development team**, and collaborated across engineering functions on **solution quality**, **release readiness**, documentation, and **production support**.
- Applied **Python**, **machine learning**, **NLP**, **statistical analysis**, **REST services**, and **full-stack development** to enterprise problems while balancing accuracy, maintainability, delivery timelines, and stakeholder needs.

RESEARCH & ENGINEERING PROJECTS

Intelligent Distributed LLM Serving Platform

- Architected a **Kubernetes**-based inference platform routing LLM requests across **eight GPU workers** using **prompt length**, **queue depth**, **latency targets**, model quality, and estimated **serving cost**.
- Implemented **continuous batching**, prefix-aware **KV-cache reuse**, **admission control**, and **load-aware scheduling**, increasing throughput **2.4x** and reducing p95 time to first token **46%** across **10,000 benchmark requests**.
- Built **Prometheus/Grafana** observability for token latency, throughput, **GPU memory**, failures, and cost; sustained **99.9% availability** during worker-failure tests and reduced cost per 1,000 tokens **31%**.

Evaluation-First Agent and RAG Platform

- Engineered a **multi-tenant LangGraph** and **FastAPI** platform combining **hybrid retrieval**, **tool execution**, **conversational memory**, **role-based permissions**, sandboxing, and **human approval** for sensitive actions.
- Developed a **250-task evaluation** framework covering retrieval quality, **groundedness**, task completion, prompt injection, unauthorized tool use, **latency**, and **cost**, achieving **91% grounded accuracy** and **87% completion**.
- Created an **uncertainty-aware controller** that retrieved additional evidence, **switched models**, **requested approval**, or **abstained**, reducing unsupported claims **63%** and unsafe tool actions **92%** versus an unrestricted baseline.

Safe Clinical Workflow Agent for Synthetic Electronic Health Records

- Built a **medication-reconciliation** support agent across **5,000 synthetic FHIR patient records**, combining **structured patient context** with evidence retrieved from **trusted clinical references**.
- Implemented **identity validation**, **evidence citation**, **uncertainty scoring**, **role-based access**, **clinician approval gates**, and **immutable audit replay** for every proposed record change.
- Achieved **94% workflow completion** and **96% evidence-grounded** recommendations across **600 simulated cases** while reducing unsafe writes **97%** and blocking **100%** of wrong-patient access tests.

Data Science Research Case Studies

- Completed **ten reproducible studies** spanning **difference-in-differences** causal inference, **CUPED** experimentation, **Kaplan-Meier** and **Cox** survival analysis, **Fourier forecasting**, and **empirical Bayes** estimation.
- Built **fraud anomaly detection** under **temporal drift**, **fairness** and **calibration** audits, **NLP ticket classification**, **spatial pollution modeling**, and **uplift modeling** with policy-value evaluation.
- Documented **data provenance**, **estimands**, **uncertainty quantification**, **diagnostics**, **robustness checks**, limitations, and executive summaries for every analysis using real and deterministic synthetic datasets.

Machine Learning Inference Notebook Collection

- Created **thirteen** self-contained notebooks for **classification**, **regression**, **forecasting**, **anomaly detection**, **sentiment analysis**, **recommendations**, and reproducible single or batch inference on local and synthetic data.
- Implemented reusable prediction interfaces with **scikit-learn pipelines** and built **K-nearest neighbors**, **gradient-descent linear regression**, and **K-means from scratch** to demonstrate algorithmic fundamentals.

Tissue-Specific Gene Expression Analysis

- Analyzed **mRNA-seq** data across liver and small-intestine tissues using **differential-expression** and **functional-enrichment** workflows to identify **tissue-specific genes**, pathways, and biological processes.
- Validated structure and biological separation through **PCA**, **multidimensional scaling**, **heatmaps**, and enrichment interpretation using standard **bioinformatics** analysis tools.

Unsupervised Deep Clustering of Single-Cell RNA-seq Data

- Built an **autoencoder**-based dimensionality-reduction workflow for **PBMC single-cell RNA-seq** data and compared **K-means**, **Leiden**, and **HDBSCAN** clustering across multiple datasets.
- Evaluated cluster structure and biological interpretability using **UMAP** visualizations and comparative **clustering diagnostics** across **PBMC 3k** and **PBMC 6k** data.

Vehicle Tracking with License Plate Recognition

- Developed a **real-time vehicle-tracking** workflow using **OpenCV** and **PyTesseract** for license-plate recognition, with **MongoDB**-backed event storage and centralized record management.
- Built an **administrative dashboard** for **route visualization**, vehicle-event lookup, and **operational monitoring** across recognized license-plate records.

TECHNICAL SKILLS

- **Programming & Querying:** Python, Java, C, C++, R, SQL, Shell scripting, JSON/JSONL
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, scikit-learn, Transformers, BERT, CNNs, autoencoders, classification, regression, clustering
- **LLM & Agentic Systems:** LangGraph, RAG, hybrid retrieval, Qdrant, tool use, memory, human-in-the-loop, vLLM, embeddings, evaluation
- **Statistics & Data Science:** Causal inference, experiment design, CUPED, survival analysis, forecasting, empirical Bayes, anomaly detection, fairness, uplift modeling
- **Backend & Applications:** FastAPI, Django, Django REST Framework, Angular, Spring Boot, REST APIs, Keycloak, HTML/CSS, JavaScript
- **Data & Databases:** PostgreSQL, MySQL, MongoDB, SQLite, Redis, data pipelines, normalization, deduplication, feature engineering
- **Infrastructure & Observability:** Docker, Kubernetes, CUDA, NVIDIA A100, Prometheus, Grafana, OpenTelemetry, MLflow, Linux, Git/GitHub
- **Biomedical & Bioinformatics:** Biomedical NLP, FHIR/HL7, HAPI FHIR, HPO, Gene Ontology, Disease Ontology, RNA-seq, Scanpy, edgeR, HISAT2, HTSeq
- **Visualization & Analytics:** UMAP, NetworkX, Matplotlib, Seaborn, PCA, Tableau, Power BI

EDUCATION

Master of Science in Bioinformatics and Data Science, University of Delaware | August 2024 - May 2026

Bachelor of Technology in Computer Science and Engineering, DKTE's Textile and Engineering Institute | August 2017 - June 2021

CERTIFICATIONS

- **Google Advanced Data Analytics Specialization:** Coursera, March 2024
  - ***Topics:*** *Foundations of Data Science; Data Analysis with R Programming; Data Visualization with Python; Machine Learning with TensorFlow on Google Cloud; Advanced SQL for Data Analysis; Predictive Modeling and Analytics; Google Advanced Data Analytics Capstone*
- **Image and Video Processing: From Mars to Hollywood with a Stop at the Hospital:** Duke University, March 2024
- **Getting Started with Power BI Desktop:** Coursera Project Network, August 2020

LEADERSHIP & AWARDS

- **President, Computer Science and Engineering Student Committee:** Led 10+ members and organized a national-level technology symposium, coding competitions, workshops, and student initiatives.
- **First Place, CODE-A-THON 2K19:** DKTE College coding competition.
- **Volunteer, Friend Youth Social Foundation:** Supported youth education, health, and community-service initiatives.